

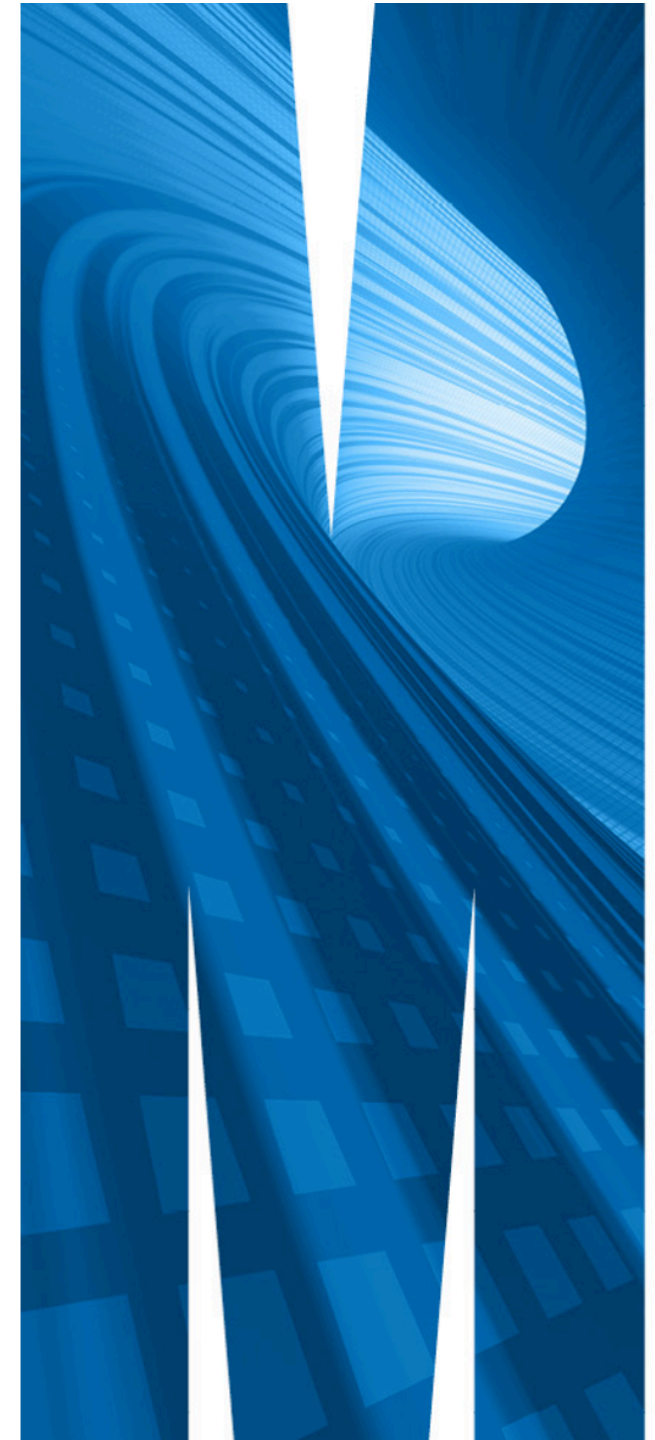


A Decade of SERJ

*Patterns, Methods, and Future Directions in
Statistics Education Research*

Alyssa Hu

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Collaborators



Abby L. Sine

Penn State University



Emilee A. Herman

San Juan College



Kaitlyn R. Fales

Root Inc.



Neil J. Hatfield

Penn State University



Matthew D. Beckman

Penn State University



Susan E. Lloyd

Bucknell University



Emi Cervantes

Penn State University



Marjorie Bond

Penn State University

Why review SERJ?

SERJ = Statistics Education Research Journal

Approaching its 25th anniversary in **2027**

Our review examines the journal's published articles, extending an earlier review.

Zieffler et al. (2011)

2002–2009 · **72 articles**

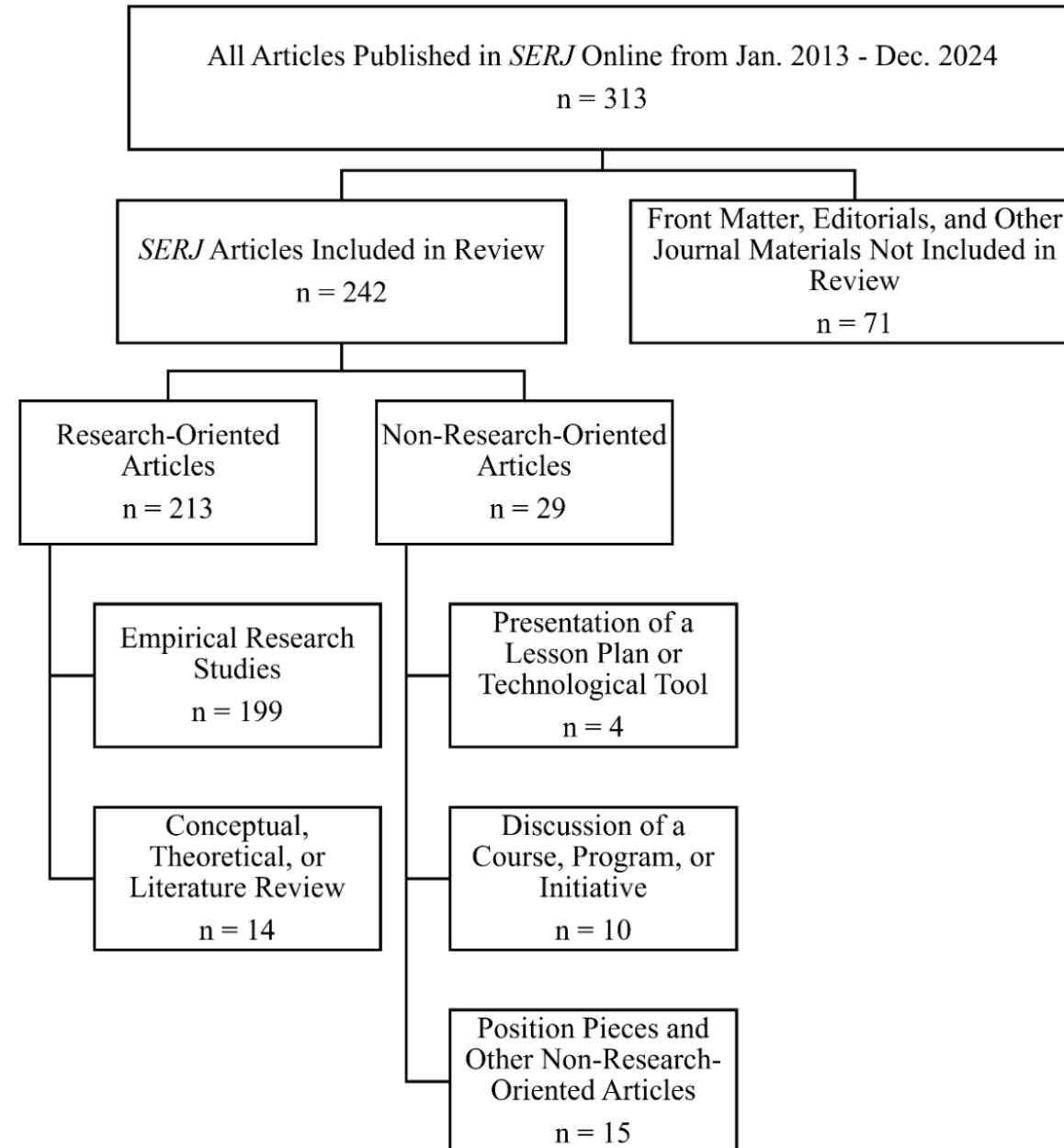


Our review

2013–2024 · **242 articles**

If you're looking for a quick tour of what statistics education research might look like, this talk is for you.

Which articles are included?



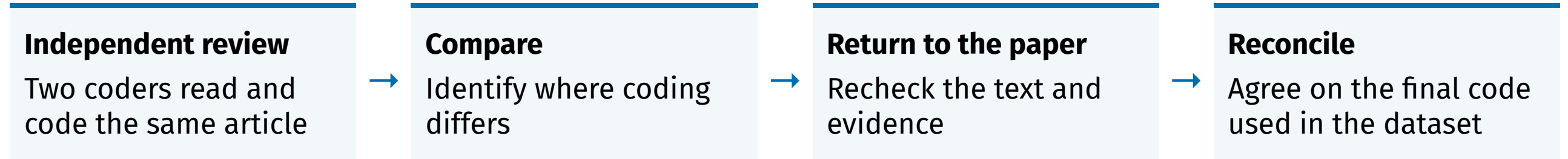
Building on the earlier review

Earlier review categories:	Data collected	Our review categories:	Data collected
Who says	Collaborations (single or multiple authors, interdisciplinary and/or international collaborations)	Authorship	Collaborations (single or multiple authors, interdisciplinary and/or international collaborations)
	Primary author(s), and the author's department, institution, country		The listed author(s) The primary affiliation listed for the author(s) (department, institution, country)
What	Research questions/goals	Research Question / Purpose	Research question(s) / aim(s) / goal(s) Research question topics
	Article topics Article keywords	Abstract and Keywords	Article abstract Article keywords

Expanding what we collect

Earlier review: question	Data collected	Our review: component	Data collected
Why	References cited	Theoretical Framework	Theoretical framework / model / perspective References pertaining to educational / learning theories
To Whom	Populations being studied	Participants	Populations being studied Participant demographics
To What Extent and What Effect	Research classification Constructs/Instruments Research methods	Methodology	Type of paper (e.g., empirical research study) Methods (quantitative, qualitative, mixed) Mechanisms for data collection Artifacts for data analysis Analysis procedures Availability of data / code / codebook
		Discussion/Study Implications	Implications (research and/or teaching)

From articles to codes



Coding form

Does the author(s) explicitly state their research question(s)/aim(s)/goal(s)/purpose(s)? (Must be able to copy/paste from the paper)

- ☐ Yes
- ☐ No
- ☐ Unsure

Written guidelines

Explicit Research Question(s)

- To answer yes to this question, you should be able to identify a specific place in the text that you can copy and paste, where the authors say what their research question, aim, goal, or purpose is. If the authors state their aims more than once, try to find the most explicit part (e.g., if they say “the aim of this study is...” and later list their research questions, copy and paste the research questions).
- If you are unsure if something counts as explicitly stated, select unsure.
- *Tip:* Look at the end of the introduction/literature review section if you can’t find anything. This is where authors will often describe the goal of the study.

Codebook and guidelines

This is an example of what the coder will see during article review.

Where are SERJ authors based?

45 countries represented · 516 unique authors · 207 unique first authors

All authors		First authors	
Country affiliation	Frequency	Country affiliation	Frequency
USA	235	USA	82
Australia	58	Brazil	23
Brazil	58	Australia	22
Germany	32	Germany	13
Canada	31	UK	10
New Zealand	26	Canada	9
UK	25	New Zealand	9
Spain	19	Spain	7
Argentina	14	Israel	7
Israel	14	Sweden	7

Across 242 articles included in review

Research in SERJ tends to be collaborative

79% of the 242 articles had multiple authors

192 multiple-author

50 single-author

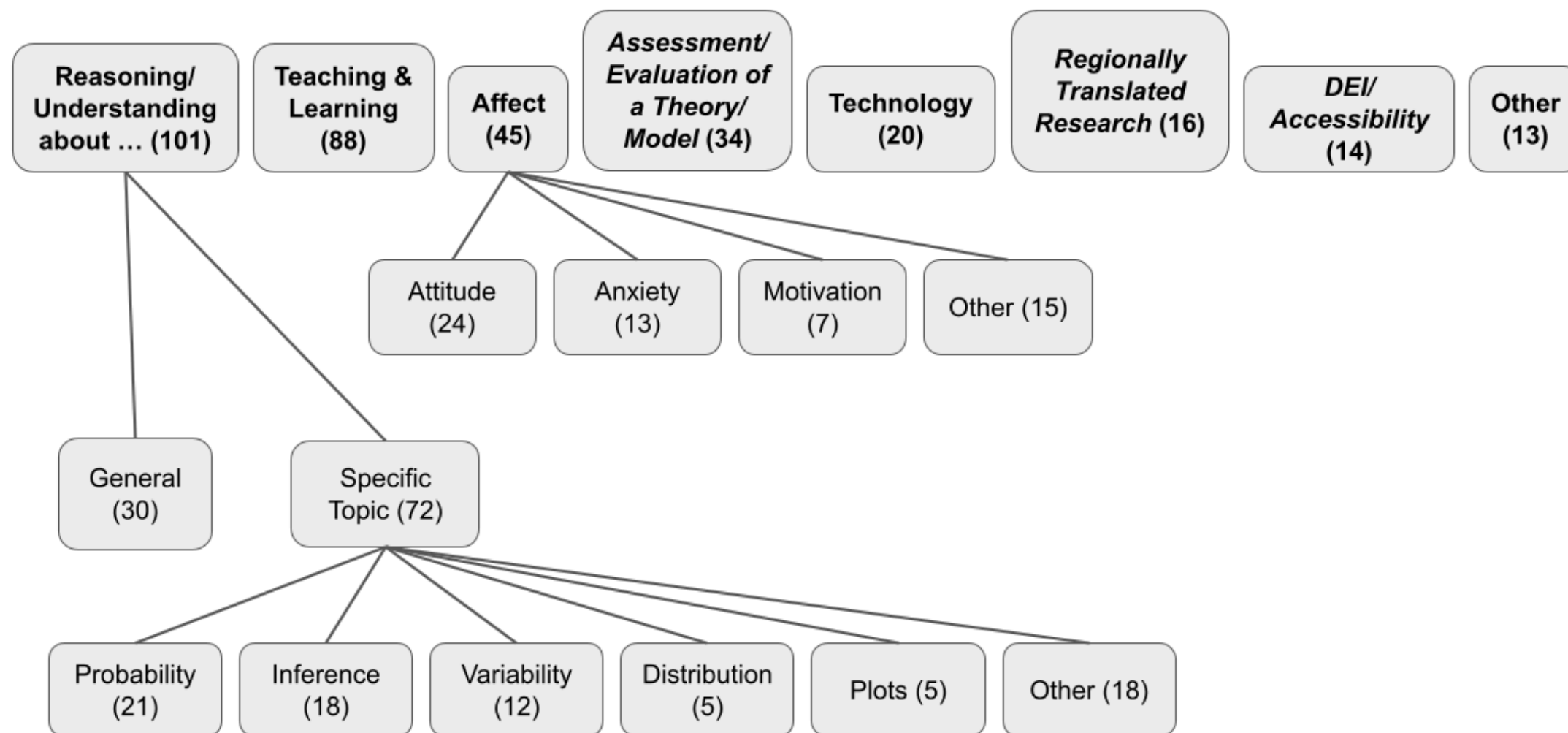
Authors per article: **mean 2.7 · median 2 · maximum 8**

111 articles involved cross-institutional collaboration

58% of the 192 multiple-author articles

43 of those 111 involved international collaboration

What topics do the articles investigate?



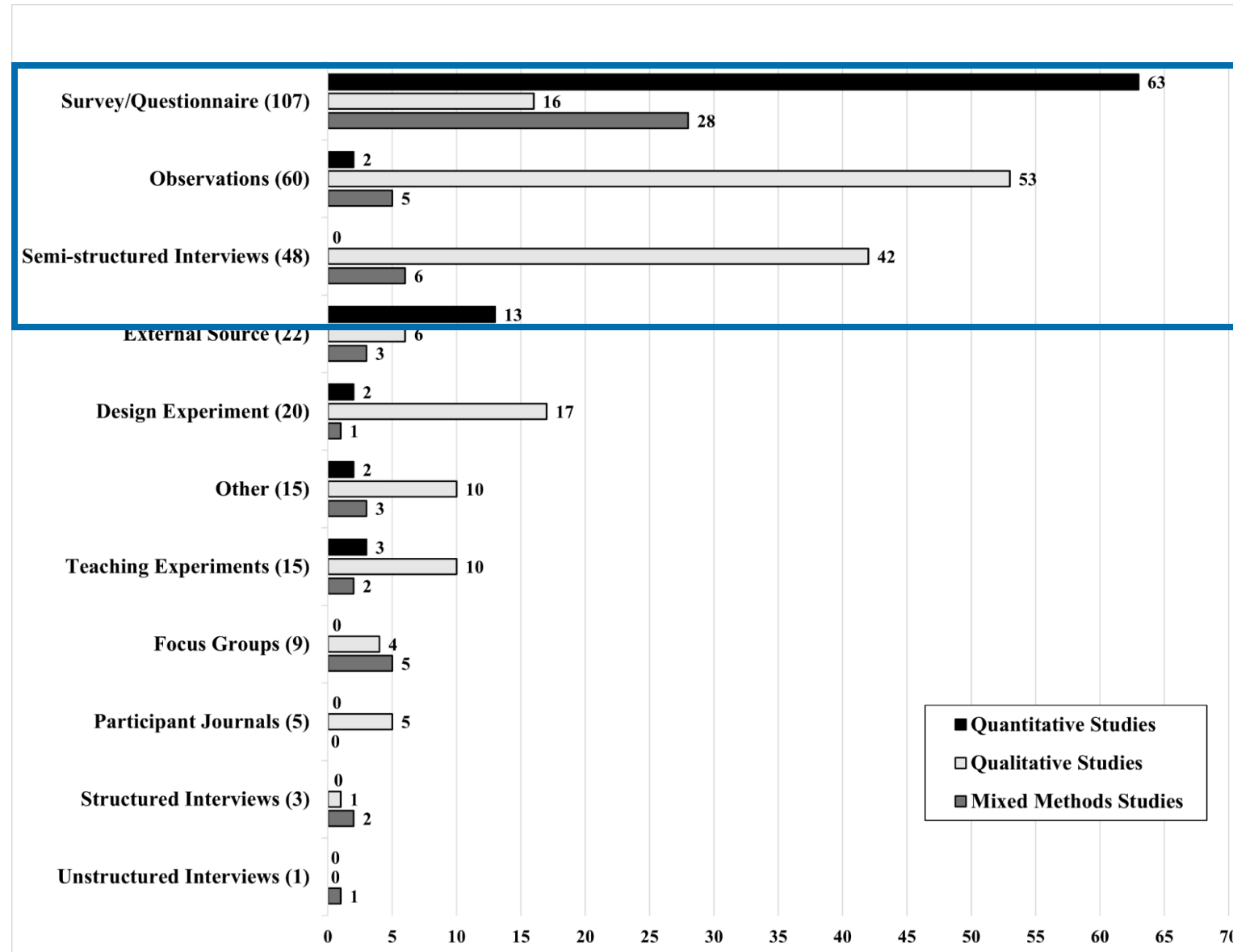
Across 213 research-oriented articles · An article may address multiple topics

Who is studied?

Population	Frequency
Undergraduate Students	76
Pre-K-12 Students	62
In-Service Teachers	23
Graduate/Professional Students	21
Pre-service Teachers	18
Literature Review	10
Post-secondary Educators	8
Textbooks, Courses, or Other Course Artifacts	7
Industry Professionals	4
Not Applicable/Theoretical	2
Non-student Adults	1
Non-traditional Students (secondary)	1

Across 213 research-oriented articles · An article may include multiple populations

How do researchers collect their data?



Across 199 empirical articles · An article may use multiple collection methods

How easy is the research to follow and build on?

Reuse and reproducibility

195 / 199 empirical articles: no data access identified

119 / 199: no study materials identified

Qualitative analysis methods were sometimes not explicitly named

Implications

196 / 213 research-oriented articles had explicit implications for teaching and/or research

Often scattered across results, discussion, limitations or conclusions

The issue was often not whether the contribution existed, but how explicitly it was communicated.

What would strengthen future research?

These directions draw on the full review, not only the selected findings shown here.

Ground the work

Clarify how theory informs research questions, design and analysis

Develop statistics-education-specific frameworks where useful

Broaden the evidence base

Attend to whose experiences and contexts are represented

Report demographics and context so readers can judge generalisability

Make the work easier to build on

Be explicit about methods and implications

Share data, instruments, materials and code where feasible

Thank you! Any questions?

Slides available at:



Preprint:

Link forthcoming